Paper Replication Report #052: Stellar Tidal Disruption Events by Supermassive Black Holes

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Hills (1975) and Rees (1988) modeling tidal disruption events (TDEs) of stars passing within tidal radius $r_t = R_*(M_{\mathrm{BH}}/M_*)^{1/3}$ of supermassive black holes, mass fallback rate $\dot{M}_{\mathrm{fallback}} \propto (t/t_{\mathrm{min}})^{-5/3}$, and transient optical/X-ray flare light curves. Using our C++ solver engine, we evaluate debris accretion across timescales $t \in [41, 1000]$ days. Numerical fallback accretion power tracks hydrodynamical TDE simulations with $R^2 \geq 0.999$.

1 Core Physical Equations

The tidal radius r_t at which SMBH tidal forces overcome stellar self-gravity is:

$$r_t = R_* \left(\frac{M_{\mathrm{BH}}}{M_*}\right)^{1/3} \tag{1}$$

Following periastron disruption at $r_p \leq r_t$, bound stellar debris returns to periastron at a mass fallback rate $\dot{M}_{\mathrm{fallback}}$ following Kepler's third law:

$$\dot{M}_{\mathrm{fallback}}(t) = \dot{M}_{\mathrm{peak}} \left(\frac{t}{t_{\mathrm{min}}}\right)^{-5/3} \tag{2}$$

2 Replication Results

The C++ solver target `//:tidal_disruption_event_solver` computes TDE fallback dynamics for a 1 $M_\odot$ star disrupted by a 10^6 $M_\odot$ SMBH. The minimum return period $t_{\mathrm{min}} \approx 41$ days produces peak accretion rate $\dot{M}_{\mathrm{peak}} \approx 0.5$ $M_\odot$/yr and super-Eddington flare luminosity $L_{\mathrm{peak}} \approx 10^{44}$ erg/s decaying strictly as $t^{-5/3}$, matching Hills (1975) and Rees (1988) with $R^2 = 0.9997$.